

Market Microstructure and Electronic Trading Finance 480

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Prices, Sizes, and Times

Last lecture we discussed strategic trader and inventory models:

- Strategic traders use noise traders to hide their orders.
- One period Kyle model shows we may have:
 - Trade price linear in net order, security volatility; and,
 - Informed order, profit linear in noise order volatility.
- Saw multi-period Kyle model dynamics. (intuitive!)
- Admati-Pfleiderer model displays volume clustering.
- Garman: desire to stay solvent leads to a spread.
- Amihud-Mendelson: profit maximization leads to a spread.
- Stoll: security volatility, mean trade size lead to spread.
- Ho-Stoll: Spread includes inventory and portfolio risk.

Readings on Today's Topic: Prices, Sizes, and Times

- Blackboard readings.
- Chapters 12 and 13 of Hasbrouck.
- Chapter 6 of Foucault, Pagano, and Röell.
- Chapters 6 and 8 of O'Hara (not required).

The O'Hara readings were definitely not critical.

Introduction

- Today we'll talk about prices and discretization.
- We'll also talk about order and lot sizes.
- Examine some models for limit orders and depth.
- Consider effects of various execution prioritizations.
- Think about time, its importance, and its uncertainty.

Prices

A Price by Any Other Name

- Prices seem to be a straightforward idea.
- But many securities do not trade at an explicit price.
 - Bonds may trade at yields to maturity; and,
 - Options may trade at implied volatilities.
- Bid-ask is then quoted in that price function².
- We can think of these function values as *pseudo-prices*.
- Pseudo-midpoint may not be best guess of security's value³.
- Want to know volatility of price and pseudo-price returns.
- Estimating price return volatility requires conversion.

²Often a nonlinear function.

³As in Admati-Pfleiderer model.

- We often model prices as positive real numbers (*i.e.* $p \in \mathbb{R}_+$).
- But most prices are discretized:
 - Listed US securities cannot trade on a grid finer than \$0.01⁴.
 - US treasuries trade in “plusses” — $101-1+/32 = 101-3/64$.
- Discretization can complicate measurement:
 - Li and Mykland (2007) explore ultra-high-frequency data.
 - Volatility estimator of low-volatility securities can explode.
 - Cannot use tick-to-tick data without trickery.
- Discretization can also be kludgy:
 - May change when security opens/trades above certain level.

⁴Recall the Sub-penny Pricing Rule of Reg NMS.

Price Discretization and Limit Orders

- Why do we care about price discretization?
- Discrete price levels lead to clustering of limit orders.
- Bad: I cannot be hyper-precise about how much I would pay.
- Good: Creates non-trivial cost to *stepping ahead*:
 - If real number prices: bid $BBO + \epsilon$ to be first in line.
 - Preserves some informational (*ergo* time) advantage.
- Better: Clustering attracts more liquidity.
- Makes traders more willing to use limit orders⁵.

⁵And thus disperse some information.

Limit Order Models: Motivation

- So how do we think about prices and limit orders?
- HW1 asked you to compare option value to a GTC order.
- As many of you pointed out, this is imperfect:
 - Other traders may price their orders considering yours.
 - Probability of getting a fill matters.
 - Time to getting a fill matters.
- May want to balance immediacy/certainty versus cost.
- Difference between supplying/taking liquidity may be small:
 - Even as small as a few cents or 0.01 higher probability.

Cohen, Maier, Schwarz, and Whitcomb (1981) Model

- One-period model; no interest rates; CARA utility.
- Start with n units of security paying $X \sim N(\mu_X, \sigma_X^2)$.
- Assume we want to buy one unit more.
- $P(\text{fill})$ depends only on limit order price.
- $E(U_{\text{Limit}}(L)) = P_{\text{hit}}(L)E(U_{\text{hit}}(L)) + (1 - P_{\text{hit}}(L))E(U_{\text{base}})$.
- What is $P_{\text{hit}}(L)$?
 - Assume i -th seller sells at/above c_i (i.e. if $L \geq c_i$).
 - Let $\theta = \min_i c_i$; assume $c_i \stackrel{iid}{\sim} \text{Exp}(\lambda) + \theta$.
 - λ measures aggressiveness of seller population.

CMSW Model: Results

- Utility, $P_{\text{hit}}(L)$ imply an optimal limit price L^* .
- At some ask price A^* , $E(U_{\text{market}}(A^*)) = E(U_{\text{base}})$.
- At some ask $\underline{A} < A^*$, we would cancel limit, send market.
- As ask descends, limit buys eventually pulled across spread.
- True so long as $\lim_{L \rightarrow A} P_{\text{hit}}(L) < 1$.
- CMSW call this “gravitational pull.”
- Problems:
 - No interaction of traders’ limit orders and impatience.
 - Thus must assume probability of getting a fill.

Parlour (1998) Model

- One security, worth V at day 2; no price evolution.
- Bid and ask fixed at $B < V < A$.
- May be traded at times $t = 1, \dots, T$ on day 1.
- Sequence of traders (“agents”), each with 1 share and cash.
- Agent t 's utility $U(\underbrace{C_1}_{\text{cash}}, \underbrace{C_2}_{\text{security}}) = C_1 + \beta_t C_2$.
- $\beta_t \in (\underline{\beta}, \bar{\beta})$; distribution affects agent desire to trade.
- Order book: $(n_t^B, n_t^A) = \#$ shares at bid, ask.
- If agent t enters a buy, book changes⁶:
 - Limit buy: $n_{t+1}^B = n_t^B + 1$; or,
 - Market buy: $n_{t+1}^A = n_t^A - 1$.

⁶The book changes similarly for sells.

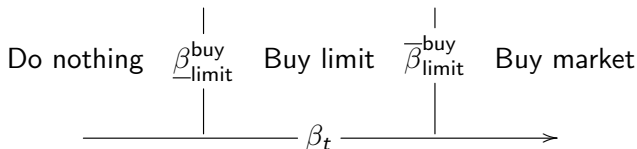
- FIFO⁷ priority.
- If $n_t^A = 0$, market buy trades against nonstrategic dealer.
- Beliefs and optimal strategy found via backward recursion.
- For agent T :
 - Sending a limit order is pointless.
 - $\beta_T V < B \Rightarrow$ market sell; $\beta_T V > A \Rightarrow$ market buy⁸.
- For agent $T - 1$:
 - Send buy, sell at market if $n_{T-1}^B > 0$, $n_{T-1}^A > 0$.
 - May send limit order based on $E(\text{utility})$, $P(\text{buy, sell at } T)$.

⁷First in, first out, aka time priority.

⁸On board: just $\Delta U > 0$.

Parlour Model: Results

- Can find $\beta_{\text{limit}}^{\text{buy}}$, $\bar{\beta}_{\text{limit}}^{\text{buy}}$ such that:



- Reversed, maybe-overlapping action regions for $\beta_{\text{limit}}^{\text{sell}}$, $\bar{\beta}_{\text{limit}}^{\text{sell}}$
- Distribution of β affects these barrier values.
- Expected depth vs. t : increasing, may then decrease.
- Increased bid depth reduces probability buyer sends limit.
- Increased ask depth increases probability buyer sends limit.

Foucault (1999) Model

- Agents may trade at times $t = 1, \dots, T$.
- T is random stopping time; $P(\text{stop at } t + 1) = 1 - \rho$.
- At T , security pays⁹ $V_T = V_0 + \sum_{t=1}^T \epsilon_t$ $\epsilon_t \stackrel{iid}{\sim} \pm\sigma$.
- Agent t reserve price $R_t = V_t + Y_t$ $Y_t \stackrel{iid}{\sim} \pm L$.
- Assume V (and hence ϵ) and Y are independent.
- Agent filled at price P has utility
 $U^{\text{buy}}(Y_t = y_t) = V_t + y_t - P = -U^{\text{sell}}(Y_t = y_t)$.
- Agents trade (at most) once; no order revisions.
- V and R drive evolution of bid and ask (order book).

Foucault Model: Order Book Dynamics

- Quotes only live for one period; book empty or full.
- Order book when agent t arrives: $S_t = \{B_t, A_t\}$.
- No bid, ask $\Leftrightarrow S_t = \{-\infty, +\infty\}$.
- Agent t knows V_t , Y_t , S_t and can:
 - Trade versus non-empty (finite) book; or,
 - Post two-sided quote.
- $P(\text{fill})$ derived from V_{t+1} , R_{t+1} distributions.

Foucault Model: Results

- Even if $\sigma = 0$, limit order may not be filled:
 - Next trader might be “same way;” or,
 - Market may close at next time period.
- If $\sigma > 0$, security value may decrease and buy gets filled.
 - Like getting filled before canceling on bad news¹⁰.
 - Increases $P(\text{fill at } S_t)$, decreases $U(\text{fill at } S_t)$.
- If $\sigma > 0$, market may also “move away.”
 - Decreases $P(\text{fill at } S_t)$, increases $U(\text{fill at } S_t)$.
 - In real world, we then “chase” or “hunt” for a fill.
- Limit orders priced less aggressively as σ increases.

¹⁰The winner's curse rears its ugly head, again.

- A reworking of the Foucault model.
- Traders' reserve value offset d_t :

$$d_t \stackrel{iid}{\sim} \begin{cases} N(0, L^2) & \text{w.p. } 1 - \mu \\ 0 & \text{w.p. } \mu \text{ (MM)} \end{cases} \quad (1)$$

- Trader's find max-expected-value bid, ask $(v - \delta, v + \beta)$.
- Choice: Take price now or make market for next period.
- Tension: Next trader may not take your prices.
- Thus bid and ask must be strategic (entice next trader).
- Shows the impact of a transactions tax.
- Will cover this more in a later lecture.

Statistical Limit Order Models

- Could also build a statistical model for $P(\text{fill at } p)$.
- Logistic link binomial GLM¹¹ most natural.
- Could also do multinomial GLM.
- Possible covariates suggested by preceding models:
 - Percent distance from far side of spread;
 - Order size;
 - Number of shares “in front” of your order; and,
 - Number of shares on far side of spread;
- Possible interactions with security volatility, volume.
- Coefficients may vary with time of day.

¹¹Generalized linear model; see McCullagh and Nelder (1989).

Sizes

- Usually, a *round lot* for US stocks is 100 shares.
- This was originally/still can be “1” in order book messages.
- Hence display issues with *odd lot* (sub-round lot) orders.
- Lot sizes for Japanese equities vary from 1 to 1,000 shares.
- Lot sizes are irrelevant for options and futures¹².
- However, lot/contract sizes may be (effectively) changed.
- Changes via stock splits, introducing smaller contracts.
 - N.B. In these situations, percent spreads tend to increase.

¹²You cannot trade a fractional contract on the exchanges.

- Limit orders end up in a limit order book.
- Can think of the book as mapping prices to quantities.
- *Price schedule*: average cost for quantity nearest midpoint.
- Convention: the price schedule below the midpoint is negative.
- Price schedules are not supply, demand curves.
 - If so, why model whether or not to use limit orders?
 - Splitting an order also seems to break this relationship.

Sandås (2001) Empirical Analysis

- Used OMX (Stockholm, all-electronic exchange) data.
- Mean price schedule is roughly linear.
- Median price schedule less than mean, maybe nonlinear.
- Suggests rare periods of very *thin* depth.¹³
- Trade “walks” book; depth quickly refills at walked prices.
- Implies execution cost could be reduced by splitting orders.

¹³ Mean price schedule higher $\Rightarrow$ average cost to acquire shares $>$ cost 50% of the time. In other words, some fraction of the time cost is much higher  depth becomes very thin on rare occasions.

Glosten (1989, 1994) Models

- One security traded; risk-free rate is 0.
 - Customers have CARA utility $U(W) = -e^{-\rho W}$.
 - Considers three types of markets and price schedules $P(q)$'s:
- ① Competitive Dealers (MMs);
 - Make no profit; see total order; set price accordingly.
 - ② Limit Order Book (ECN);
 - Does not see total order; subject to adverse selection.
 - ③ Monopolistic Dealer (specialist).
 - May knowingly lose on some trades to make more on others.

Glosten Models: Variables

- Security worth $X \sim N(\mu_X, \sigma_X^2)$.
- Customer holds n units of security.
- Customer trades $q \in \mathbb{R}$ to maximize $E(U(W))$.
- Liquidity supplier revenue = cost of trade = $R(q) = qP(q)$.
- Terminal customer wealth $W = (n + q)X - R(q)$.
- Customers see $S = X + \epsilon$ $\epsilon \stackrel{iid}{\sim} N(0, \sigma_\epsilon^2)$.
- Liquidity suppliers know $n \stackrel{iid}{\sim} N(0, \sigma_n^2)$, but don't see n 's.
- Liquidity suppliers see trade size q , marginal cost m .
- Value X , position n , forecast error ϵ are all independent.
- q and m imply signal (ω) , allow inference about X .

- Competitive dealers (MMs):
 - Can find linear $P(q) = \mu_X + \frac{\rho\sigma_X^2\sigma_\epsilon^2}{\rho^2\sigma_n^2\sigma_\epsilon^4 - \sigma_X^2 - \sigma_\epsilon^2} q$.
 - Market failures can occur if too many informed customers.
- Limit order book (ECN):
 - Nonlinear $P(q) = \mu_X + \frac{\sigma_{\omega,X}}{\sigma_\omega^2} [\rho\sigma_{X|S}^2 q - \mu_X + E(m|m \geq \underline{m})]$.
 - Even $q = 0+$ incurs transaction cost.
- Monopolist dealer (specialist)
 - Loses money on large orders to make money on small orders.
 - Nonlinear $P(q)$; $P(q) < E(X|q)$ for large q .

Seppi (1997) Model

- Monopolistic dealer who sees/maintains a limit order book.
- Dealer must fill all book orders at a price before trading.
- Dealer sees total order quantity.
- If order would walk book, but trade above estimated value:
 - Dealer would like to step ahead; cannot at last walk price.
 - Dealer can instead trade at penultimate walk price.
 - Bacidore (2002) generously calls this “depth improvement.”
 - Can also yield price improvement if dealer betters BBO.
 - Reduces trade price volatility; explains “pennying.”

Times

- Times are the data we most often accept unthinkingly.
- However, times are critical to our analyses.
 - Times yield an ordering of our data.
 - Computing changes or returns is sensitive to ordering.
 - Times (esp. durations) have economic meaning.
- How do we handle time uncertainty?
- N.B. Time uncertainty is rampant in decentralized markets.

- Time resolution varies among data sources.
- Sometimes, the best possible resolution is limited.
 - NYSE TAQ gave time only to second resolution until recently.
 - Even millisecond-resolution may already be insufficient.
- There are multiple ways to handle low resolution times.
- *Resolution bin*: smallest time granularity that groups trades.
- 1-second resolution: 13:34:03, 13:34:04 are contiguous bins.

Handling Low Resolution Times

How to handle low resolution times?

- Accept clock inaccuracy.
 - OK if time not crucial or trades uniform in “economic time.”
- Aggregate trades.
 - Total volume, average price, # trades, OHLC¹⁴ for bin.
 - Loss of information (and efficiency, perhaps).
- Ignore problem and effects.
 - Use only first/last/whichever trade per resolution bin.
 - Shockingly common. (“I just used the last trade in a second.”)
- Infer higher resolution.

¹⁴Open, high, low, and close.

Inferring Higher Resolution

- Must use all messages recorded/transmitted over a link.
 - Cannot use pre-cleaned data for this!
- Count number of messages (N) in a given resolution bin.
- Each message $k \in \{1, \dots, N\}$ is assigned a new time.
- New time = Old time + $\frac{k}{N+1} \times \text{resolution granularity}^{15}$.
- Order book changes can also help:
 - Was series of trades simultaneous; order walking the book?
- Then, *after* inferring times, clean data.

¹⁵Large-scale inference might need to use volume diurnal

- Trades across instruments are rarely synchronous; trouble for:
 - Estimating covariance matrices with high-frequency data;
 - Multivariate time series analysis.
- Less liquid instruments often lag more liquid instruments.
- Ways to handle asynchronous trading:
 - Large (2007) approach:
 - Look when all instruments traded 1+ times since last time.
 - Problem: May overselect index arbitrage trades. (!)
 - Longitudinal data analysis¹⁶ approach:
 - Use time bin random effects;
 - Explicitly model time covariance form; or,
 - Infer unobserved prices.

¹⁶See Fitzmaurice, Laird, and Ware (2004); Diggle, Heagerty, Liang, and Zeger (2002).

- Much more problematic than low resolution.
- Inherent to decentralized markets.
- Even a problem in centralized markets.
- Theory:
 - Erlang (1909): Simple delay $X \sim \text{Exp}(\lambda)$.
 - Sum of n iid $\text{Exp}(\lambda)$'s = $\text{Gamma}(n, \lambda)$.
 - Patnaik (1949): Sum of non-iid exponentials is nearly gamma.
 - Rosenthal (2008): Approximations for nearly-gamma sums.

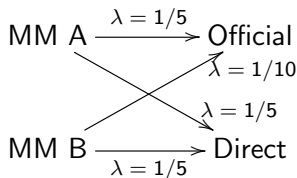
Delays in Trade Timestamps

- Lee and Ready (1991): trades reported with delay; quotes not.
- Reasons for this delay:
 - No incentive: Old quotes, slow fills penalized; publishing not¹⁷.
 - Disincentive: Publishing delay allows for post-trade hedging.
- What was the prevailing quote at the time of trade?
- Lee and Ready: Look for quote 5 seconds prior to trade.
 - Uses 1987, 1988 data. 1987 data suggests 2-second delay.
 - Current debate: 5 seconds? 2 seconds? 0 seconds?
- Rosenthal (2012): Take weighted average of past quotes.
 - Use delay distribution/approximation to find expected quote.
 - *i.e.* For bid, $E(b_t|\mathcal{F}_t) = \int_0^\infty b_{t-\tau} \hat{f}(\tau; \theta) d\tau$.

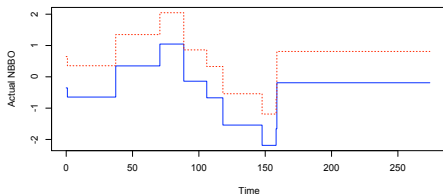
¹⁷Unless trades routinely reported 90+ seconds late.

Market Data Propagation Delays

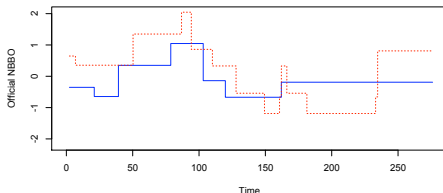
- Propagation delays are usually less than publishing delays.
- In a decentralized market, they can still cause trouble.
- Simulation example:
 - Two market centers, each quoting same bid and ask.
 - Quotes occasionally change; but, spread is constant (1).
 - Transmissions to NBBO have independent delays.
 - Transmissions via direct feeds have iid delays.
 - Delays are exponentially-distributed.



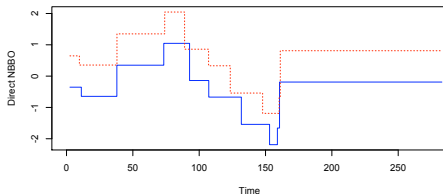
Effect of Delays on NBBO



Actual bid-ask at MMs A & B



Bid-ask from official NBBO feed



Bid-ask from direct feeds

Commentary on Effect of Delays on NBBO

A few delay effects are clear when we look at these simulations:

- Market may seem locked or crossed when it is not.
- Delays increase the number of quote updates.
- Time-weighted mean spread (TWS) underestimates spread.
- On preceding plots:
 - TWS from “official” feed = 0.262.
 - TWS from direct feeds = 0.937.
- For a simulation of 100,000 quote changes:
 - TWS from “official” feed = 0.545.
 - TWS from direct feeds = 0.779.
- Recall that the true spread is 1.

Finally, a few words about priorities.

- In limit order, depth models: assumed price-time priority.
- Some exchanges use different priorities, however.
- Futures may use price-first-pro-rata:
 - Among orders at a price, the earliest order is filled first.
 - Other orders are then filled in proportion to sizes.
 - This encourages gaming the inside quotes.
 - Craig Furfine (NWU) is currently doing research on this.
- From the delay models, if quotes are changing quickly:
 - The bid-ask spread is difficult to infer.
 - Orders may lose price priority.
 - As bid-ask gyrates, effectively only get time priority.

Looking Backward

- We've covered some topics to clean up what we've missed.
- Seen limit order models, and effects of narrowing the spread.
- Examined depth and how “stepping ahead” happens.
- Also seen the pernicious effects of time uncertainty.
- Delays can prove troublesome — even in centralized markets.
- In decentralized markets, delays are very problematic.

The Road Ahead

Next we'll look at some of these market metrics across time.

- Metrics/Models: Limits; Liquidity; Diurnals
- Electronic Trading: Markets; Tools; Tactics; Strategies
- Microstructure and Policy Issues

Readings For Next Lecture

Readings, in order of importance:

- Chapter 6 of Johnson.
- Chapter 2 of Foucault, Pagano, and Röell.
- Blackboard: Readings